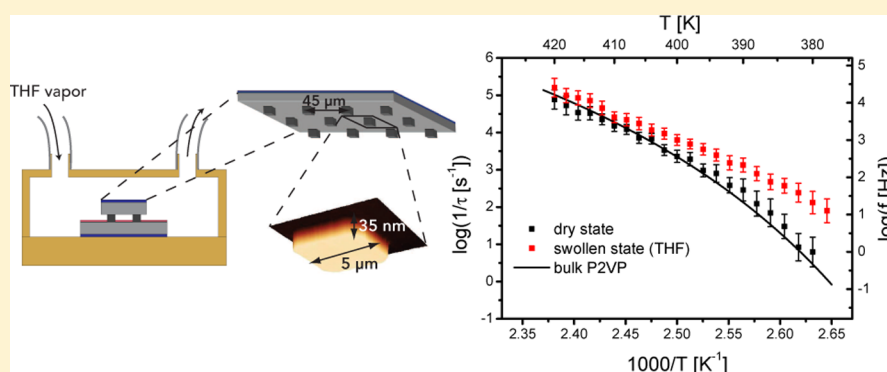


## Molecular Dynamics of Swollen Poly(2-vinylpyridine) Brushes

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## S Supporting Information



**ABSTRACT:** The influence of swelling on the molecular dynamics of poly(2-vinylpyridine) (P2VP) brushes is measured by broadband dielectric spectroscopy (BDS) in a broad temperature (350–420 K) and spectral (0.1 Hz–1 MHz) range with nanostructured, highly conductive silicon electrodes, separated by 35 nm high insulating silica spacers. A “grafting-to” method is applied to prepare P2VP brushes with a grafting density  $\sigma = 0.12 \text{ nm}^{-2}$  and a film thickness  $d = 7.3 \text{ nm}$  as measured by ellipsometry. Swelling of the P2VP brushes is realized with tetrahydrofuran (THF) vapor using a flow cell. In the dry state, the segmental dynamics of the P2VP brushes coincides with the dynamic glass transition of the bulk system while in the swollen state it becomes faster by up to 1–2 decades due to the plasticizing effect of THF.

## ■ INTRODUCTION

There is a broad range of interests on polymer brushes with respect to fundamental and applied research. Different synthesis strategies have been established for various materials with the goal to tune surface properties by design of the tethered polymer layer resulting in various types of systems like mixed brushes and composites with nanoparticles or bifunctional polymers.<sup>1–11</sup>

Beside applications in the field of corrosion, wetting, or cell adhesion, thin polymer layers and polymer brushes are interesting from a more fundamental point of view. The confinement of polymers in layers of only a few nanometers thickness and further restrictions like grafting of polymers in the case of polymer brushes raised the question of whether properties like the glass transition and glassy dynamics deviate from bulk properties.<sup>12,13</sup> It has been shown that the calorimetric glass transition may deviate from the bulk value.<sup>14–21</sup> The segmental motion or  $\alpha$ -relaxation is the structural relaxation responsible for the dynamic glass transition. For thin films<sup>17,22–29</sup> and isolated polymer coils<sup>30</sup> it has been shown that glassy dynamics is identical to bulk dynamics. Also for polymer brushes with end-grafted chains

and different grafting densities no deviation from bulk dynamics could be measured, which shows that the intermolecular potentials determining glassy dynamics are not altered.<sup>31</sup>

External stimuli like pH value, temperature, or solvents are used to swell thin polymer films. In multicomponent systems like block copolymers or mixed polymer brushes, the selective swelling of a particular component can be used to switch between different states and induce e.g. changes in the surface properties. Many experiments studied structural changes in the morphology and the effects on the surface properties.<sup>32–44</sup> In contrast, the present work addresses the question of how the molecular dynamics, i.e., glassy dynamics, of a polymer brush is affected by swelling with tetrahydrofuran (THF), a good solvent for P2VP. Since each P2VP chain segment carries a dipole moment, broadband dielectric spectroscopy (BDS) allows for the measurement of the segmental relaxation, i.e., the dynamic glass transition. An approach based on nanostructured electrodes is used,<sup>45</sup> where silica nanostructures with

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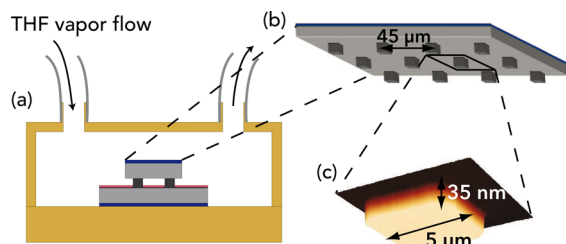
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a height of 35 nm serve as spacers. This enables dielectric measurements of ultrathin films having a free upper surface being accessible to solvent vapor molecules.

## EXPERIMENTAL SECTION

Two ultraflat, highly conductive silicon wafers (Si-Mat), with a specific resistance  $\rho < 0.02 \Omega \text{ cm}$  and a 200 nm thick Al-layer on the backside, serve as electrodes for the sample capacitor. The lower electrode, a  $10 \times 20 \text{ mm}^2$  silicon wafer die, is used as substrate while the top electrode, a  $1 \times 1 \text{ mm}^2$  silicon wafer die, is equipped with a regular matrix of highly insulating silica nanostructures. They have a height of 35 nm and a base area of  $5 \times 5 \mu\text{m}^2$ , serving as spacers to guarantee well-separated electrodes (Figure 1). Consequently, the thin film



**Figure 1.** Sample arrangement: (a) measurement flow cell with a sample capacitor and a controllable THF vapor flow, which enables dielectric measurements of THF swollen P2VP brushes. The capacitor is composed of two highly doped silicon electrodes (bottom electrode:  $10 \times 20 \text{ mm}^2$ ; top electrode:  $1 \times 1 \text{ mm}^2$ ), where the top electrode is covered by an array of insulating silica nanostructures, serving as spacers (b). The spacers have a height of 35 nm, as confirmed by AFM scans (c) to allow dielectric measurements of thin films with a free upper surface.

inside the capacitor has a free upper surface without a (possibly evaporated metal) electrode influencing the thin film. For the brush preparation a “grafting-to” procedure in two steps is applied.<sup>7,31,46–48</sup> After cleaning the wafer in an ultrasonic bath with ethanol and subsequent oxygen plasma treatment, poly(glycidyl methacrylate) (PGMA) ( $M_n = 17.5 \text{ kg/mol}$ ,  $M_w = 29.7 \text{ kg/mol}$ , Polymer Source Inc.) is spin-coated from a 0.02 wt % chloroform solution ( $d \approx 2.5 \text{ nm}$ ). The polymer brush is formed by carboxyl-terminated poly(2-vinylpyridine) (P2VP-COOH) ( $M_n = 40.6 \text{ kg/mol}$ ,  $M_w = 43.8 \text{ kg/mol}$ , Polymer Source Inc.) which is deposited from a 1 wt % THF solution followed by 16 h annealing and finally extraction in THF. During the

preparation, the film thicknesses are measured by ellipsometry (Sentech SE-402 scanning microfocus ellipsometer).

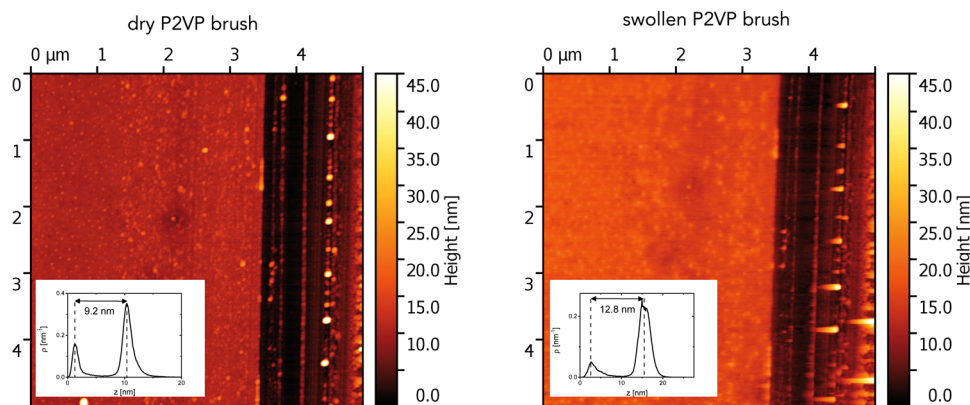
Atomic force microscopy (AFM) using a Veeco Dimension 3000 AFM in tapping mode is carried out to measure the height change of the P2VP brush after swelling. With a sharp razor blade a scratch is made in the thin film, and the height profile is measured first in the dry state and then in the swollen state. To examine the swollen film thickness, the sample is exposed to THF vapor directly in the AFM setup by using a home-built box surrounding the AFM with vapor inlet and outlet. The box is connected with a THF vessel. Nitrogen gas, flowing through the vessel with THF, is used to control the THF vapor flow.

To allow for dielectric measurements of the P2VP brush in the dry and swollen state, a flow cell is used (Figure 1) with electrical connections for the sample capacitor (not depicted) and an inlet and outlet for THF vapor. Similar to the setup used for the AFM measurements, the flow cell is connected with a THF vessel outside the cryostat, and the vapor flow is controlled by nitrogen gas, flowing through the vessel with THF. For the dielectric measurements, a Novocontrol Alpha analyzer having a spectral range of 1 mHz–10 MHz and a resolution in  $\tan \delta$  of  $10^{-4}$  is used. Temperature is controlled by a Quatro temperature controller (350–420 K) with 0.1 K relative accuracy.

## RESULTS AND DISCUSSION

During preparation, the thicknesses of the deposited layers of PGMA and P2VP are measured by ellipsometry. The PGMA anchoring layer has a thickness of  $d_{\text{PGMA}} = 2.3 \pm 0.2 \text{ nm}$  while the measured P2VP brush layer thickness is  $d_{\text{P2VP}} = 7.3 \pm 0.2 \text{ nm}$ . With this ellipsometrically measured P2VP layer thickness the grafting density of the brush can be calculated using  $\sigma_{\text{P2VP}} = \rho N_A d_{\text{P2VP}} / M_n$  ( $N_A$ : Avogadro constant).<sup>49</sup> Inserting the molecular weight  $M_n = 40.6 \text{ kg/mol}$  and the density  $\rho = 1.14 \text{ g/cm}^3$ , the grafting density is  $\sigma_{\text{P2VP}} = 0.12 \pm 0.01 \text{ nm}^{-2}$  with a corresponding distance between the grafting points of about 3.3 nm. This is smaller than the radius of gyration of P2VP ( $R_g \approx 5 \text{ nm}$ ), and the grafted P2VP layer is in the “true-brush” regime.<sup>49</sup>

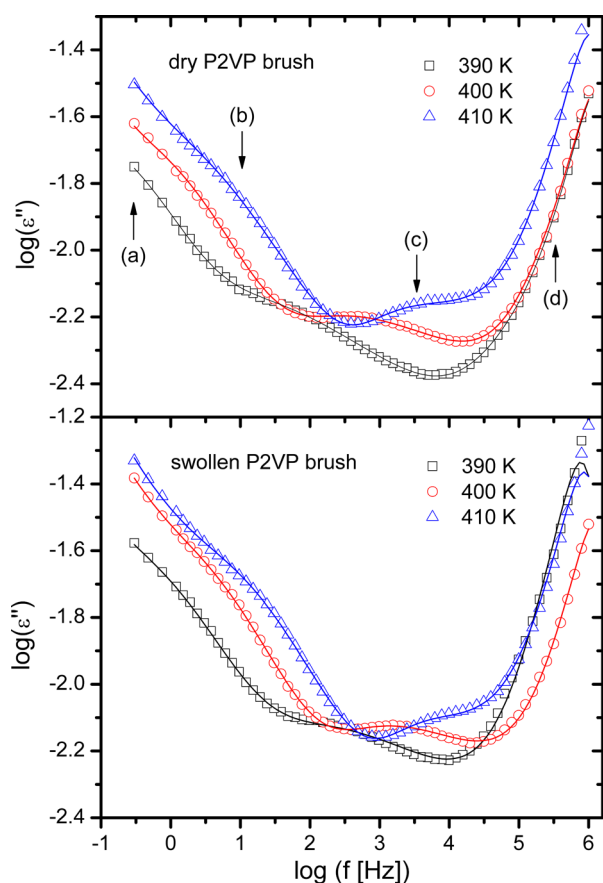
Swelling of the P2VP brush results in a height increase of the thin film. This height change is measured in situ by AFM while exposing the sample to THF vapor. Typical images of the surface topography with the corresponding height distributions are given in Figure 2. In the dry state a layer thickness  $d_{\text{dry}} = 9.5 \pm 1 \text{ nm}$  is measured. This value comprises both layers of the P2VP and PGMA since they cannot be distinguished. The



**Figure 2.** AFM scans to check the film thickness of the P2VP brushes in the dry state and after swelling in THF vapor. The P2VP brushes were prepared with a grafting density of  $\sigma_{\text{P2VP}} = 0.12 \pm 0.01 \text{ nm}^{-2}$  and a dry film thickness of  $d_{\text{PGMA}} = 2.3 \pm 0.2 \text{ nm}$  and  $d_{\text{P2VP}} = 7.3 \pm 0.2 \text{ nm}$  measured by ellipsometry. A scratch in the film is used to determine the film thickness. The images show example scans in the dry and the swollen state. The average total film thickness for six scans is in the dry state  $d_{\text{dry}} = 9.5 \pm 1 \text{ nm}$  and after swelling  $d_{\text{swollen}} = 12.4 \pm 1 \text{ nm}$ . This reveals a mean swelling factor  $d_{\text{swollen}}/d_{\text{dry}} = 1.3 \pm 0.2$ .

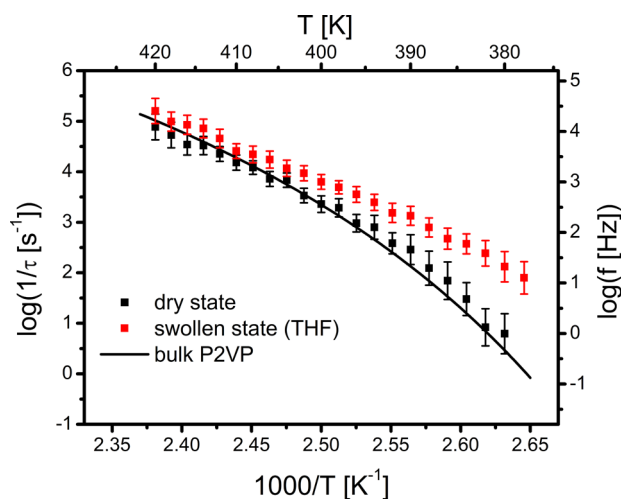
larger uncertainties are caused by local inhomogeneities which are not captured by the ellipsometry measurements. In the swollen state a film thickness  $d_{\text{swollen}} = 12.4 \pm 1$  nm is measured. This gives a mean swelling factor of  $d_{\text{swollen}}/d_{\text{dry}} = 1.3 \pm 0.2$ .

The influence of the solvent vapor and the respective swelling of the P2VP brushes on the segmental dynamics is measured by BDS in the frequency range from 0.1 Hz to 1 MHz and at temperatures between 350 and 420 K. The spectra are recorded in terms of the complex dielectric function  $\epsilon^* = \epsilon' - i\epsilon''$ , especially using the imaginary or loss part ( $\epsilon''$ ). The dipolar fluctuations and the corresponding molecular relaxation processes are best represented by this quantity. In Figure 3, spectra of P2VP brushes in the dry state and after swelling in THF vapor are displayed for three temperatures (390–410 K). Beside the P2VP and PGMA molecular relaxations, the silicon



**Figure 3.** Dielectric loss  $\epsilon''$  spectra of a P2VP brush ( $\sigma_{\text{P2VP}} = 0.12 \text{ nm}^{-2}$ ) on a PGMA anchoring layer before and after swelling in THF vapor at varying temperatures as indicated. Several contributions are observed and marked by arrows in the spectra of the dry brush: (a) a conductivity contribution caused by the spacers or contaminating particles between the electrodes, (b) a polarization due to the conductivity of P2VP, (c) the P2VP  $\alpha$ -relaxation, and (d) a contribution generated by the limited electrical conductivity within the material of the electrodes. The PGMA relaxation in the measured samples is covered by the electrode conductivity contribution on the high frequency side and is only visible as a wing in the spectra at the given temperatures. An equivalent circuit model<sup>31,50</sup> including complex dielectric functions of all the components present in the sample capacitor is used to derive a fit function (solid lines). In particular, the P2VP and PGMA relaxations are described by two Havriliak–Negami functions. The experimental error corresponds to the symbol size.

electrodes and the silica spacers contribute to the spectra. The employed fit function (Figure 3, solid lines) includes all components present in the capacitor with their specific dielectric functions and is based on an equivalent circuit model.<sup>31,50</sup> A detailed description of the equivalent circuit and the fit function as well as an overview of the values of the characteristic parameters of the relaxations resulting from the fits can be found in the Supporting Information. A parasitic conductivity caused by contaminating particles or imperfect spacers leads to a contribution on the low frequency side. A further contribution on the high frequency side is caused by the limited conductivity of the wafers. Also inside the P2VP layer a conductivity is present and causes a polarization peak (marked as (b) in Figure 3). Two Havriliak–Negami functions  $\epsilon^* = \epsilon_{\infty} + \Delta\epsilon/(1 + (i\omega\tau_{\text{HN}})^{\beta})^{\gamma}$ , an empirical function to describe relaxation processes,<sup>51</sup> are used to fit the P2VP and PGMA relaxations. As described earlier,<sup>31</sup> the PGMA contribution is observable in a temperature range from 200 to 420 K. At the given temperatures in Figure 3, the PGMA relaxation is covered by the high frequency resistivity contribution of the wafers and only present as a wing, leading to high uncertainties of the PGMA relaxation times. A more detailed discussion of the extracted relaxation times of PGMA is given in the Supporting Information. The relaxation process marked with (c) in Figure 3 is identified with the P2VP  $\alpha$ -relaxation. Comparing the spectra of the dry and the swollen state at a single temperature, the frequency of the maximum position of the P2VP  $\alpha$ -relaxation shifts to higher values. This maximum indicates the mean relaxation rate, which in the dry state follows a Vogel–Fulcher–Tammann<sup>52–54</sup> temperature dependence similar to the glassy dynamics of bulk P2VP (Figure 4). This result has been reported earlier<sup>31</sup> and indicates that the intermolecular potentials between the chain segments are not affected by the stretching of the chains in the brush, suggesting a bulklike density in these brush systems. In contrast, in the swollen P2VP



**Figure 4.** Activation plot of the P2VP  $\alpha$ -relaxation of a P2VP brush ( $\sigma_{\text{P2VP}} = 0.12 \text{ nm}^{-2}$ ) in the dry state and swollen by THF vapor. The dry P2VP brush (black squares) follows a Vogel–Fulcher–Tammann temperature dependence similar to the mean relaxation rate of bulk P2VP (solid line). In the swollen state, the temperature dependence of the mean relaxation rate is shifted (red squares) due to a solvent-induced enhancement of the P2VP  $\alpha$ -relaxation. The error bars show the accuracy for determining the mean relaxation rate as a parameter from the Havriliak–Negami fits.



brush a clear shift of the  $\alpha$ -relaxation to higher frequencies compared to bulk dynamics by up to 1–2 decades can be observed (Figure 4) in a temperature range of 380–405 K. This result can be understood considering the effect of the solvent on the molecular dynamics in the swollen state. It acts as plasticizer<sup>55–57</sup> in the P2VP brush. An increase of free volume due to the presence of the solvent results in faster molecular dynamics of the polymer chains, i.e., an increase of the mean relaxation rate. The fact that the relaxation rates at temperatures above 405 K coincide for the dry and swollen state point to the possibility that the solvent uptake above temperatures around 405 K is negligible. Compared to earlier studies, investigating structural changes of brushes exposed to external stimuli, this result shows that molecular dynamics change as well and differ from bulklike glassy dynamics.

## CONCLUSION

P2VP polymer brushes in the dry state and swollen by THF were investigated by dielectric spectroscopy to study the effect of swelling on the molecular dynamics. Layer thicknesses were measured in the dry state during preparation by ellipsometry and by AFM in the dry and swollen state to reveal the degree of swelling. The measurement of the dynamic glass transition of swollen P2VP brushes was possible using a capacitor made of nanostructured electrodes with 35 nm silica spacers combined with a flow cell in the spectrometer. Compared to the dry state with bulklike behavior of the mean relaxation rate, the molecular dynamics in the swollen state clearly differs from bulk. The presence of the solvent changes the intermolecular potentials and the relaxation rate increases. For the first time, the solvent-induced enhancement of the relaxation rate, a well-known plasticizer effect, has been proven experimentally for a monolayer of polymer brushes. Beyond conformational changes, external stimuli like solvents can also be used to alter molecular dynamics, which represents another mechanism to manipulate the properties of polymer brushes.

## ASSOCIATED CONTENT

### Supporting Information

The Supporting Information is available free of charge on the ACS Publications website at DOI: 10.1021/acs.macromol.6b00363.

A detailed description of the equivalent circuit model and the deduced fit function for the analysis of the dielectric spectra; an overview of the values of the fit parameters for the selected spectra displayed in Figure 3; details of the extracted shape parameters of P2VP and the mean relaxation times of PGMA (PDF)

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### Notes

The authors declare no competing financial interest.

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